Comparative Evaluation of Vision-Language Models for Metal Surface Defect Classification

Expert Prompting and 4-bit QLoRA Adaptation of Three Architecture-Distinct VLMs on the GC10-DET Dataset

Shixiang Wang, Yongjing Wang, Richard Fox

School of Electronic, Electrical and Systems Engineering (EESE), University of Birmingham

Research question: Under a shared dataset split, Expert Prompting strategy and 4-bit QLoRA framework, how do three architecture-distinct open-source VLMs compare in predictive performance, class-wise behaviour and deployment feasibility?

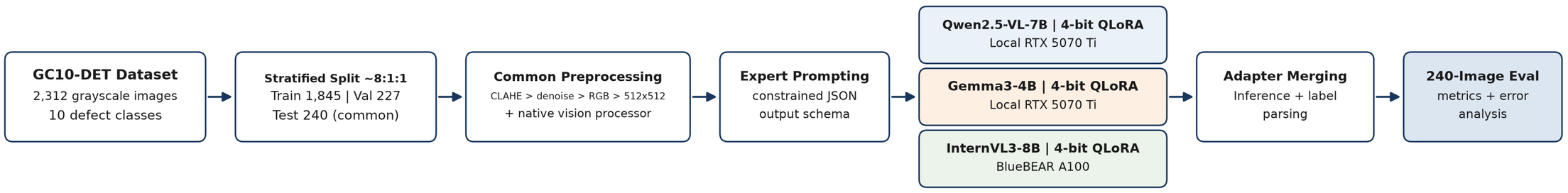


Figure 1. Experimental workflow. All three VLMs were adapted with 4-bit QLoRA in LLaMA-Factory on a shared stratified split and evaluated on the identical 240-image test set; each model retained its native vision processor, so adaptation settings remained partly model-specific.

1. Background

Automated metal-surface inspection is essential for industrial quality control, yet conventional vision models need task-specific architectures for a fixed class set. Vision-language models (VLMs) offer a unified framework for image understanding and instruction-based classification, but general benchmarks do not reflect specialised industrial imagery: metal defects show low contrast, subtle texture change and strong inter-class similarity, combined with substantial class imbalance.

Three open-source VLMs were selected along the architectural spectrum rather than by popularity, so the comparison covers distinct design strategies under one task, one split, one prompting strategy and one fine-tuning framework.

2. Objectives

- Adapt three representative VLMs to ten-class defect classification with 4-bit QLoRA.

- Compare overall and class-wise performance on a shared held-out test set.

- Identify shared and model-specific directional misclassification patterns.

- Examine deployment constraints across a consumer workstation and an HPC platform.

- Assess whether a smaller VLM offers a competitive accuracy-resource trade-off.

3. Dataset

The image-level experimental dataset was derived from the local GC10-DET release and contains 2,312 grayscale steel-surface images in ten defect categories; images were organised into mutually exclusive class folders, giving a single-label classification task. A stratified split (seed 42) produced 1,845 training, 227 validation and 240 test images (approx. 8:1:1); the 240-image test set is identical for all models.

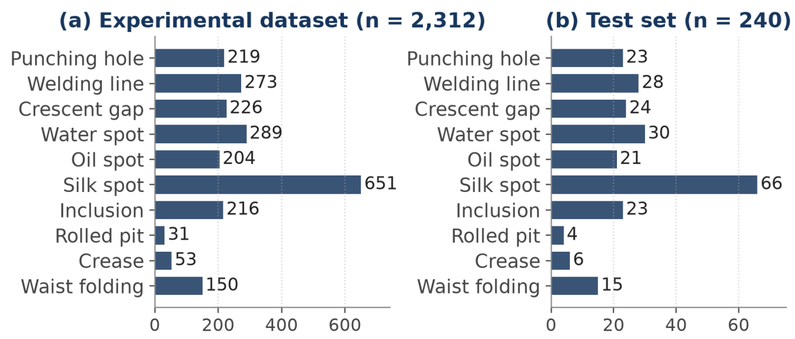


Figure 2. Class distribution of the experimental dataset and the held-out test set. Silk spot accounts for 28.2% of the dataset while rolled pit accounts for 1.3%; macro-averaged metrics are therefore reported alongside accuracy.

4. Example Defect Images

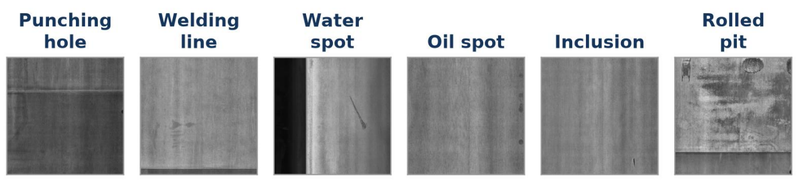


Figure 3. Preprocessed GC10-DET examples (CLAHE-enhanced, 512 x 512). Visually similar spot-like defects (water spot, oil spot, inclusion) form the dominant confusion group.

5. Supporting View: Confusion Matrices

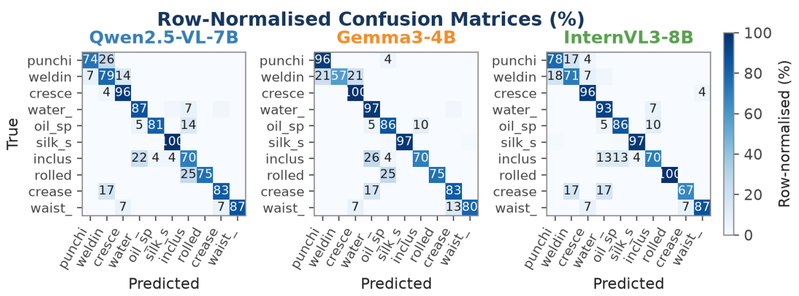


Figure 4. Row-normalised confusion matrices (%): diagonal dominance is comparable across models while off-diagonal mass concentrates in the two confusion groups of Section 9.

6. Architecture-Driven Model Selection

Model	Architecture rationale
Qwen2.5-VL-7B	Dynamic-resolution ViT + M-RoPE; fine-grained visual features
Gemma3-4B	SigLIP encoder, compressed visual tokens; resource-conscious route
InternVL3-8B	Native multimodal pre-training; dynamic high-res tiling
LLaVA-OneVision-7B	SigLIP + MLP projector + AnyRes; excluded (integration)

7. Methodology

Image preprocessing. A common reference pipeline converted each image to grayscale, enhanced local contrast with CLAHE (clip limit 2.0, 8 x 8 tile grid), applied light Gaussian denoising (3 x 3 kernel, sigma 0.8), converted to three-channel RGB and resized to 512 x 512; each model then applied its native vision processor, so the final visual-token representation remained model-specific.

Expert Prompting (EP). Each model acted as a senior metal-surface inspection expert, received per-class morphological descriptions and a three-step analysis scaffold, and returned a strict JSON object. Only the defect_type field was used for quantitative evaluation; auxiliary severity and description fields were excluded because GC10-DET provides no ground-truth severity annotations.

Prompt (abridged): You are a senior metal-surface quality-inspection expert. Domain knowledge: ten defect classes with morphology descriptions (e.g. punching_hole: round/oval holes; welding_line: linear weld traces; inclusion: embedded foreign particles). Steps: (1) inspect for anomalous regions; (2) match the class descriptions; (3) determine defect type and severity. Output strictly JSON only: {"defect_type": <class>, "severity": <mild|moderate|severe>, "description": <brief morphology>}.

Training. All three VLMs were adapted with 4-bit NF4 QLoRA in LLaMA-Factory: LoRA rank 16, alpha 32, dropout 0.05 on the language-model projections (q/k/v/o/gate/up/down); 3 epochs; LR 1e-4 with cosine schedule and 20-step warm-up; batch 1 with gradient accumulation 8; cutoff 1024; bf16 with gradient checkpointing; seed 42. Vision towers were not trained in any model; for InternVL3-8B the vision tower and multimodal projector were additionally frozen explicitly. Adapters were merged into the base weights before evaluation.

Evaluation. Labels were parsed from the JSON defect_type field and scored on the shared test set with accuracy, Macro-F1, Weighted-F1 and per-class precision, recall and F1.

8. Overall Predictive Performance

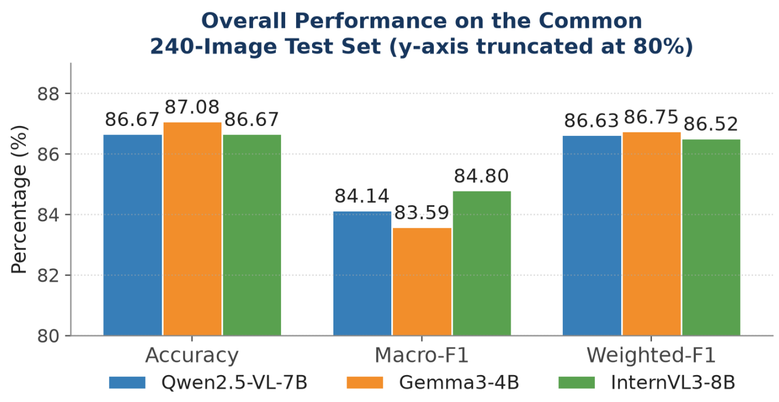


Figure 5. Overall performance of the three QLoRA-adapted VLMs (vertical axis truncated at 80%). Per model: Qwen2.5-VL-7B Acc 86.67 / Ma-P 86.19 / Ma-R 83.05 / Ma-F1 84.14 / W-F1 86.63, 32 errors; Gemma3-4B 87.08 / 85.99 / 84.00 / 83.59 / 86.75, 31 errors; InternVL3-8B 86.67 / 85.57 / 84.44 / 84.80 / 86.52, 32 errors. Macro metrics recomputed from the multiclass confusion matrices; weighted metrics follow class support.

Key result: accuracy is closely matched (86.67-87.08%; gap 0.41 percentage points, i.e. one test image), yet the accuracy and Macro-F1 rankings disagree, so aggregate accuracy alone cannot rank models on this imbalanced test set.

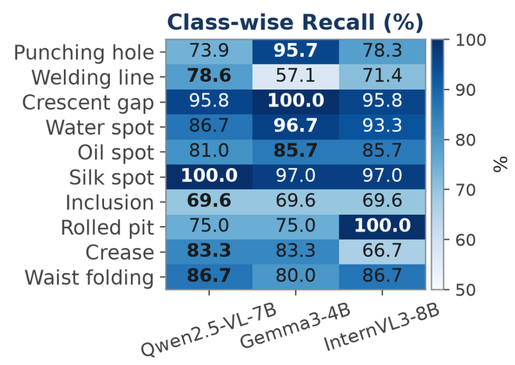


Figure 6. Class-wise recall. Gemma3-4B led on punching hole, crescent gap and water spot; Qwen2.5-VL-7B reached perfect recall on silk spot; InternVL3-8B classified all four rolled-pit samples correctly. Inclusion recall was 69.57% for all three models; rolled pit (n=4) and crease (n=6) should be read cautiously.

9. Per-Class F1 and Error Analysis

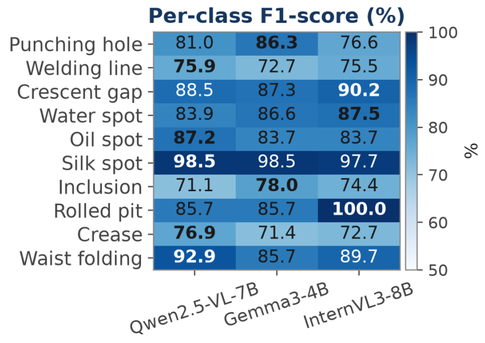


Figure 7. Per-class F1-score. No model dominated every class: Qwen2.5-VL-7B led on oil spot, silk spot, crease and waist folding; Gemma3-4B on punching hole and inclusion; InternVL3-8B on crescent gap, water spot and rolled pit.

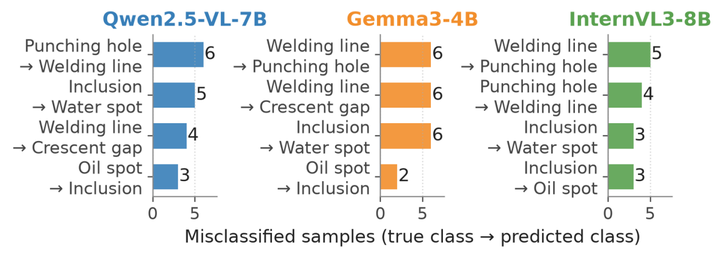


Figure 8. Dominant directional misclassifications per model (true class to predicted class).

Two recurrent confusion groups appear: punching hole versus welding line, and inclusion versus spot-like defects. Both members of each group can produce strong local edge or contrast cues, and inclusions repeatedly predicted as spots show that small, low-contrast features remain difficult after adaptation.

Why are the confusion matrices asymmetric? Entries Cij and Cji are different class-conditional errors and are not theoretically equal; unequal class support magnifies asymmetry in raw counts, while directional visual ambiguity sets the dominant error flow.

10. Deployment Feasibility

Model	Environment	Latency	Observation
Qwen2.5-VL-7B (7B)	Local RTX 5070 Ti	1.690 s/img	Locally executable
Gemma3-4B (4B)	Local RTX 5070 Ti	2.529 s/img	Locally executable
InternVL3-8B (8B)	BlueBEAR A100	62.723 s/img	HPC after local load failure

Latency is environment-specific and not directly comparable. InternVL3-8B failed to load on the local Windows workstation (bitsandbytes 4-bit error; BF16 load exceeded the page-file limit) but completed normally on BlueBEAR A100 - a platform-compatibility issue, not an inherent HPC requirement.

Gemma3-4B: highest accuracy with the smallest parameter count on a consumer GPU.

11. Excluded Baseline and Prior Research

LLaVA-OneVision-7B was planned as a fourth family (SigLIP encoder, projector alignment, AnyRes multi-tile) but excluded after persistent AnyRes token-alignment failures under the BlueBEAR storage quota - an integration limitation, not evidence of poor ability. Expert Prompting previously improved VLM defect classification on the six-class NEU benchmark (+12.7% accuracy for Qwen2.5-VL-7B) [8]; our prompt design was informed by it. LNG-tank research independently evaluated Qwen-, Gemma-3- and LLaVA-family models under resource-constrained QLoRA [9]; direct numerical comparison with our setting is inappropriate.

12. Key Findings and Conclusions

Headline numbers: 87.08% highest accuracy (Gemma3-4B); 84.80% highest Macro-F1 (InternVL3-8B); 86.19% highest Macro precision (Qwen2.5-VL-7B); 0.41 pp maximum accuracy gap (one test image); 69.57% inclusion recall, all three models.

- Three architecture-distinct VLMs matched closely overall under a shared EP and 4-bit QLoRA framework.
- Gemma3-4B: highest accuracy (87.08%) and Weighted-F1 (86.75%) with fewest parameters; InternVL3-8B: highest Macro-F1 (84.80%) and Macro recall (84.44%).

- Similar accuracy concealed large class-specific differences; recall, F1 and directional errors must be read jointly.
- Inclusion is a shared weakness; punching hole and welding line form the main reciprocal confusion group; deployment observations are platform-specific.

13. Limitations and Reproducibility

- Test set limited to 240 images; rolled pit (n=4) and crease (n=6) metrics are unstable; the 0.41-pp gap equals one image.
- No ablation of CLAHE/denoising or of the EP schema; native processors yield different visual tokens.
- No confidence scores, hence no ROC/PR curves or paired tests; latency from heterogeneous platforms.
- Code, prompt template, evaluation scripts and supplementary tables will be released on GitHub.

References

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